AIM:Support Vector Machines (SVM)
Implement the SVM algorithm for binary classification.
Train an SVM model using a given dataset and optimize its parameters.
Evaluate the performance of the SVM model on test data and analyze the results.

CODE:

```
import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns
from sklearn.metrics import classification_report, confusion_matrix
from sklearn.model_selection import GridSearchCV, train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC


df = pd.read_csv("C:/Users/hrane/Downloads/online food delivery_107.csv")
df.columns = df.columns.str.strip()
df = df.apply(lambda x: x.str.strip() if x.dtype == "object" else x)


df["Output"] = df["Output"].map({"Yes": 1, "No": 0})
X = pd.get_dummies(
    df.drop(["Output", "Feedback"], axis=1, errors="ignore"), drop_first=True
)
y = df["Output"]

X_scaled = StandardScaler().fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.2, random_state=42, stratify=y
)

param_grid = {"C": [0.1, 1, 10], "kernel": ["linear", "rbf"]}
grid = GridSearchCV(SVC(random_state=42), param_grid, cv=3, scoring="accuracy")
grid.fit(X_train, y_train)


best_model = grid.best_estimator_
y_pred = best_model.predict(X_test)

print(f"Optimized Parameters: {grid.best_params_}")
print(classification_report(y_test, y_pred))


sns.heatmap(confusion_matrix(y_test, y_pred), annot=True, fmt="d", cmap="Blues")
plt.title("SVM Classification Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
```


RIYA T107

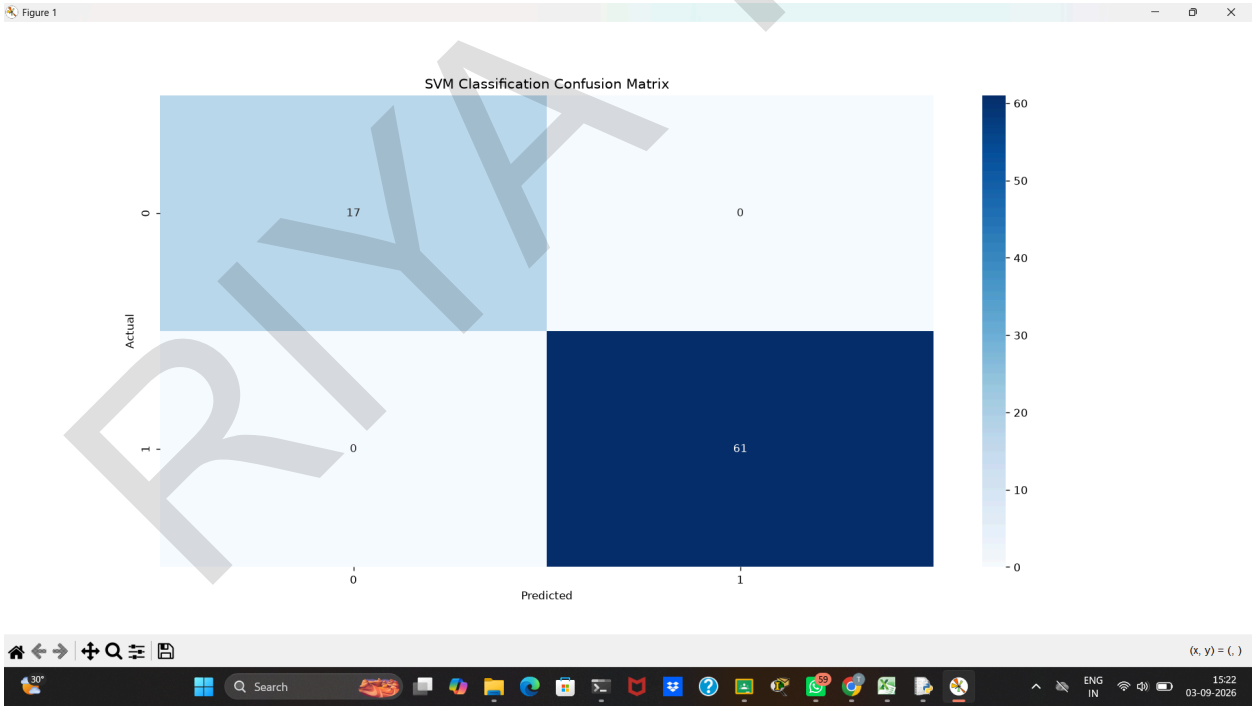
plt.show()

OUTPUT:

```
*IDLE Shell 3.14.7*
File  Edit  Shell  Debug  Options  Window  Help
Python 3.14.7 (tags/v3.14.7:823f032, Aug  5 2026, 10:51:32) [MSC v.1944 64 bit
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:\Users\hrane\Documents\AI_R.py ==========
Optimized Parameters: {'C': 0.1, 'kernel': 'linear'}
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        17
           1       1.00      1.00      1.00        61

    accuracy                           1.00        78
   macro avg       1.00      1.00      1.00        78
weighted avg       1.00      1.00      1.00        78
```



RIYA T107